
Sensor (encoder)
Raw IQ X
RD FFT and spectral tiles
Z
Pruning, shared mask
Dithered quantization, step $\widehat{\sigma}_b 2^{6-q_b}$
Entropy coding
12 iterations
ρ_b
q_b
R
CA-CFAR pass on X
Signal, CFAR, and allocation features
PPO actor, price λ
Budget rule $R \le R_{\max}$
Rate-limited link
$R \le R_{\max}$
Side information: $\widehat{\sigma}_b, \rho_b, q_b, P_b$, mask
Remote processor (decoder)
Entropy decoding, dequantization
Noise filling, power P_b
Inverse FFT
Reconstructed $\widehat{X}$
RD processing and detection (CA-CFAR or network)
Training only:
not used at run time
PPO update
Per-tile reward r_b from CA-CFAR outcomes: labels, robust detections, false peaks, rate, error
CA-CFAR on X, labels, B_b, ε_b